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**UMK65FDKSPC00000OA Foundry Design Kit of 65
nm Logic and Mixed-Mode Standard Performance
Low-K Process**

(Ver. B04_P. B)

**DSM NO. : G-9FD-LOGIC/MIXED_MODE65N-SP/LOW_K/UMK65FDK
SPC00000OA-FDK**

Total Pages : 14 (INCLUDING THIS COVER PAGE)

Approve Date : 2016/05/30

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2. Revision History

Ver.	Phase	Approved Date	From	To	Remark (Purpose)
B04	B	05/30/2016	-	1. Upgrade DRC command file and technology file 2. Modify the source/drain pin sequence of all mixed-mode MOS PCells to be forward-compatible to Virtuoso 6.1.6. 3. Add the (nil m 1) parameter mapping to calview.cellmap files for Calibre XRC back-annotation 4. Change "parseAsNumber" attribute of matalOption parameter in CDF to no for mixed-mode MOMCAP/MIMCAP/PAD_RF/metal resistor	

3. Design Rule Check

A. UMK65FDKSPC000000OA_B04_PB.tar.gz

1. stream.map - directory containing the Cadence stream in and out maps
2. cds.lib - file containing the Cadence initialization file.
3. lib.defs - file containing the Cadence initialization file.
4. display.drf - ASCII version of display resources file.
5. Models - directory containing the model files used to test the FDK.
6. umc65sp.tf - ASCII version of technology file.
7. umc65sp – FDK Technology Library.
8. RuleDecks – directory containing the Assura rule decks.
9. .pdkRepository - Files for different IO voltage
10. CDL_include_file.cir - Include file for CDL netlist

B. doc

FDK_OA_User_Guide_V1_5.pdf

Release_Note_UMK65FDKSPC000000OA_B04_PB.pdf

FDK_Application_Note_UMK65FDKSPC000000OA_B04_PB.pdf

QA_Report_G-9FD-LOGIC_MIXED_MODE65N-SP_LOW_K-UMK65FDKSPC000000OA-FDK-B04_PB.pdf

Table 1. Foundry Process Documents DSM List

Classification	Spec. No.	Version	Date
Design Support Manual	G-01-LOGIC/MIXED_MODE65N-SP/LOW_K-DSM	1.6_P1	2011/09/08
Electrical Design Rule	G-02-LOGIC/MIXED_MODE65N-SP/LOW_K-EDR	1.7_P2	2013/01/25
Topological Layout Rule	G-03-LOGIC/MIXED_MODE65N-SP-TLR	1.12_P1	2015/05/05
Interconnect Capacitance	G-04-LOGIC/MIXED_MODE65N-LOW_K-INTERCAP	1.2_P2	2016/04/15
SPICE Modeling	G-05-LOGIC/MIXED_MODE65N-SP-SPICE	1.3_P2	2011/06/21
	G-05HP-MIXED_MODE/RFCMOS65N-MIM/LOW_K-SPICE/HSPICE/CAPACITOR	1.0_P1	2008/12/08
	G-05HM-RFCMOS65N-1.0V-SP_RVT-SPICE/HSPICE/MOSFET	1.0_P1	2009/06/08
	G-05HM-RFCMOS65N-1.8V-SP-SPICE/HSPICE/MOSFET	0.2_P1	2008/05/30
	G-05HM-RFCMOS65N-2.5V-SP-SPICE/HSPICE/MOSFET	1.0_P1	2009/12/09
	G-05HM-RFCMOS65N-3.3V-SP-SPICE/HSPICE/MOSFET	1.0_P1	2009/06/12
	G-05HL-RFCMOS65N-OIF/LOW_K-SPICE/HSPICE/INDUCTOR	1.3_P1	2011/06/21
	G-05HO-RFCMOS65N-LOW_K-SPICE/HSPICE/PAD	1.0_P1	2008/10/09
	G-05HP-RFCMOS65N-1.0V_1.1V-SP/MIS_VAR-SPICE/HSPICE/CAPACITOR	1.0_P1	2009/06/19
	G-05HP-RFCMOS65N-1.8V-SP/MIS_VAR-SPICE/HSPICE/CAPACITOR	0.2_P1	2008/03/05
	G-05HP-RFCMOS65N-2.5V-SP/MIS_VAR-SPICE/HSPICE/CAPACITOR	1.0_P1	2009/06/08
	G-05HP-RFCMOS65N-3.3V-SP/MIS_VAR-SPICE/HSPICE/CAPACITOR	1.0_P1	2010/02/05
	G-05HP-RFCMOS65N-MOM/LOW_K-SPICE/HSPICE/CAPACITOR	1.0_P1	2009/04/22
	G-05HR-RFCMOS65N-SP-SPICE/HSPICE/RESISTOR	T.0_P1	2008/02/13
	G-05SP-MIXED_MODE/RFCMOS65N-MIM/LOW_K-SPICE/SPECTRE/CAPACITOR	1.0_P1	2008/12/08
	G-05SP-MIXED_MODE/RFCMOS65N-MIM/LOW_K-SPICE/SPECTRE/CAPACITOR	1.0_P1	2008/12/08
	G-05SR-RFCMOS65N-SP-SPICE/SPECTRE/RESISTOR	T.0_P1	2008/02/13

Classification	Spec. No.	Version	Date
SPICE Modeling	G-05SM-RFCMOS65N-1.0V-SP_RVT-SPICE/SPECTRE/MOSFET	1.0_P1	2009/06/08
	G-05SM-RFCMOS65N-1.8V-SP-SPICE/SPECTRE/MOSFET	0.2_P1	2008/05/30
	G-05SM-RFCMOS65N-2.5V-SP-SPICE/SPECTRE/MOSFET	1.0_P1	2009/12/09
	G-05SM-RFCMOS65N-3.3V-SP-SPICE/SPECTRE/MOSFET	1.0_P1	2009/06/12
	G-05SL-RFCMOS65N-OIF/LOW_K-SPICE/SPECTRE/INDUCTOR	1.3_P1	2011/06/21
	G-05SO-RFCMOS65N-LOW_K-SPICE/SPECTRE/PAD	1.0_P1	2008/10/09
	G-05SP-RFCMOS65N-1.0V_1.1V-SP/MIS_VAR-SPICE/SPECTRE/CAPACITOR	1.0_P1	2009/06/19
	G-05SP-RFCMOS65N-1.8V-SP/MIS_VAR-SPICE/SPECTRE/CAPACITOR	0.2_P1	2008/03/05
	G-05SP-RFCMOS65N-2.5V-SP/MIS_VAR-SPICE/SPECTRE/CAPACITOR	1.0_P1	2009/06/08
	G-05SP-RFCMOS65N-3.3V-SP/MIS_VAR-SPICE/SPECTRE/CAPACITOR	1.0_P1	2010/02/05
	G-05SP-RFCMOS65N-MOM/LOW_K-SPICE/SPECTRE/CAPACITOR	1.0_P1	2009/04/22
Mask Tooling	G-06-LOGIC/MIXED_MODE65N-SP-MASKTOOL-12A	1.7_P1	2012/08/17
DRC Rule Deck	G-DF-LOGIC/MIXED_MODE65N-SP/CALIBRE-DRC	1.12_P1	2015/06/03
LVS Rule Deck	G-DF-LOGIC/MIXED_MODE65N-SP/LOW_K/CALIBRE-LVS	1.7_P1	2013/11/08
Layer Map	G-DF-GENERATION65N-VIRTUOSO-TF	1.10_P1	2015/11/13

Classification	EDA Tool	Version
Schematic Entry	Cadence Composer	IC6.1.5.500.1
Simulation Interface	Cadence Analog Design Environment	IC6.1.5.500.1
Simulation Tool	Cadence Spectre	Please refer to SPICE
	Synopsys Hspice	Please refer to SPICE
Layout Editor	Cadence Virtuoso	IC6.1.5.500.1
DRC Tool	Mentor Calibre	Please refer to DRC deck
LVS Tool	Mentor Calibre	Please refer to LVS deck

Device Type	Device Name
MOS	N_11_SPSHVT
MOS	N_BPW_11_SPSHVT
MOS	P_11_SPSHVT
MOS	N_11_SPRVT
MOS	N_BPW_11_SPRVT
MOS	P_11_SPRVT
MOS	N_11_SPLVT
MOS	N_BPW_11_SPLVT
MOS	P_11_SPLVT
MOS	N_11_SPHVT
MOS	N_BPW_11_SPHVT
MOS	P_11_SPHVT
MOS	N_11_SPNVT
MOS	N_18_SPNVT
MOS	N_25_SPNVT
MOS	N_33_SPNVT
MOS	N_25OD33_SPNVT
MOS	N_18_SP
MOS	P_18_SP
MOS	N_33_SP
MOS	N_33_SP
MOS	N_BPW_33_SP
MOS	P_33_SP
MOS	N_25UD18_SP
MOS	N_BPW_25UD18_SP
MOS	P_25UD18_SP
MOS	N_25_SP
MOS	N_BPW_25_SP
MOS	P_25_SP
MOS	N_25OD33_SP
MOS	N_BPW_25OD33_SP
MOS	P_25OD33_SP
MOS	N_10_SPRVTRF
MOS	P_10_SPRVTRF

Device Type	Device Name
MOS	N_BPW_10_SPRVTRF
MOS	P_18_SPRF
MOS	N_18_SPRF
MOS	P_25_SPRF
MOS	N_25_SPRF
MOS	P_33_SPRF
MOS	N_33_SPRF
MOS	N_BPW_18_SPRF
MOS	N_BPW_25_SPRF
MOS	N_BPW_33_SPRF
BJT	PNP_V32X32_SP
BJT	PNP_V20X20_SP
BJT	PNP_V50X50_SP
BJT	PNP_V100X100_SP
BJT	NPN_V32X32_SP
BJT	NPN_V20X20_SP
BJT	NPN_V50X50_SP
BJT	NPN_V100X100_SP
Diode	DION_SP
Diode	DIOP_SP
Diode	DIONW_SP
Diode	DIODP_SP
Diode	DIODNW_SP
CAP	NCAP_11_SP
CAP	NCAP_18_SP
CAP	NCAP_25_SP
CAP	NCAP_33_SP
CAP	MOMCAPS_SY_MMKF
CAP	MOMCAPS_AS_MMKF
CAP	MOMCAPS_SYMESH_MMKF
CAP	MOMCAPS_ASMESH_MMKF
CAP	MIMCAPS_20F_NWELL_RFKF
CAP	MIMCAPS_20F_PSUB_RFKF

Device Type	Device Name
CAP	MIMCAPS_20F_MM
CAP	MIMCAPS_20F_M1_RFKF
CAP	MOMCAPS_ARRAY_VP3_RFVCL
CAP	MOMCAPS_ARRAY_VP4_RFVCL
CAP	MOMCAPS_ARRAY_VP5_RFVCL
CAP	VARMIS_11_RF
CAP	VARMIS_18_RF
CAP	VARMIS_25_SPRF
CAP	VARMIS_33_SPRF
RES	RNPD_SP
RES	RNND_SP
RES	RNPPO_SP
RES	RNNPO_SP
RES	RNHR_SP
RES	RSPD_SP
RES	RSND_SP
RES	RSPPO_SP
RES	RSNPO_SP
RES	RSNWELL_SP
RES	R_M1_SP**
RES	R_1X_SP**
RES	R_2X_SP**
RES	R_4X_SP**
RES	R_6X_SP**
RES	RNHR_SPRF
RES	RNHR_NW_SPRF
RES	RNNPO_SPRF
RES	RNNPO_NW_SPRF
RES	RNPPO_SPRF
RES	RNPPO_NW_SPRF
IND	L_SLCR30K_RFVIL
IND	L_SQSK_RFVIL
IND	L_SY30K_RFVIL

Device Type	Device Name
IND	L_SYCT30K_RFVIL
IND	L_SLCR30K_RFVIL_1P6M1T1F1U
IND	L_SQSK_RFVIL_1P6M1T1F1U
IND	L_SY30K_RFVIL_1P6M1T1F1U
IND	L_SYCT30K_RFVIL_1P6M1T1F1U
IND	L_SLCR8K_RFVIL
IND	L_SY8K_RFVIL
IND	L_SYCT8K_RFVIL
PAD	PAD_RF

4. Appendix Revision History

Ver.	Phase	Approved Date	From	To	Remark (Purpose)
B01	B	09/04/2007	-	-	Original
B02	B	11/12/2010	-	1. Add devices: NCAP_33_SP, N_33_SPNVT, N_33_SP, P_33_SP, RNHR_SP, N_25OD33_SPNVT,, N_25_OD33_SP, P_25OD33_SP, DIODP_SP, DIODNW_SP, PNP_V20X20_SP, PNP_V50X50_SP, PNP_V100X100_SP, NPN_V20X20_SP, NPN_V50X50_SP, NPN_V100X100_SP, NPN_V32X32_SP, MOMCAPS_SYMMESH_MMKF, MOMCAPS_ASMESH_MMKF, N_10_SPRVTRF, P_10_SPRVTRF, N_25_SPRF, P_25_SPRF, N_BPW_10_SPRVTRF, N_BPW_25_SPRF, VARMIS_11_RF, VARMIS_18_RF, VARMIS_25_SPRF, L_SLCR30K_RFVIL, L_SY30K_RFVIL, L_SYCT30K_RFVIL, L_SLCR8K_RFVIL, L_SY8K_RFVIL, L_SYCT8K_RFVIL, L_SQSK_RFVIL, N_18_SPRF, P_18_SPRF, N_BPW_18_SPRVTRF, PAD_RF, RSPD_SP, RSND_SP, RSPPO_SP, RSNPO_SP, RNHR_SP, N_25UD18_SP, P_25UD18_SP, N_11_SPSHVT, P_11_SPSHVT, RSNWELL_SP, Metal Resistor (R_M1_SP, R_1X_SP, R_2X_SP, R_4X_SP, R_6X_SP), RNPPPO_SPRF, RNPPPO_NW_SPRF,	

Ver.	Phase	Approved Date	From	To	Remark (Purpose)
B02	B	11/12/2010	-	RNNPO_SPRF, RNNPO_NW_SPRF, RNHR_SPRF, RNHR_NW_SPRF, MOMCAPS_ARRAY_VP3_RFVCL, MOMCAPS_ARRAY_VP4_RFVCL, MIMCAPS_20F_PSUB_RFKF, MIMCAPS_20F_M1_RFKF, MIMCAPS_20F_NWELL_RFKF, MIMCAPS_20F_MM, N_BPW_11_SPHVT, N_BPW_11_SPRVT, N_BPW_11_SPSHVT, N_BPW_11_SPLVT, N_BPW_25_SP, N_BPW_25UD18_SP, N_BPW_25OD33_SP, N_BPW_33_SP, N_33_SPRF, N_BPW_33_SPRF, P_33_SPRF, MOMCAPS_ARRAY_VP5_RFVCL, VARMIS_33_SPRF 2. Add metal options: 3, 6, 9, 14, 15, 18, 20, 21, 22, 25, 26, 27, 30, 74, 76, 77, 78 3. Change metal option from 1P9M0T2H0U_option1 to 1P9M0T2H0U_option79 4. Remove metal option 1P9M0T2F0U_option2 5. Remove Die seal ring generator in UMC utility 6. Revised the Pins of "Plus" and "Minus" ports for "MOMCAPS_AS_MMKF" to align with model document 7. Modified Pcell: 7-1 Add MMSYMBOL layer for MOMCAPS stacking devices:N_11_SPRVT, P_11_SPRVT, N_11_SPLVT, P_11_SPLVT, N_11_SPHVT, P_11_SPHVT, N_11_SPNVT, N_18_SPNVT, N_25_SPNVT, N_18_SP, P_18_SP, N_25_SP, P_25_SP, PNP_V32X32_SP, DION_SP, DIOP_SP, DIONW_SP, RNPDP_SP, RNND_SP, RNPPPO_SP, RNNPO_SP 7-2 Add layers MMSYMBOL, RFSYMBOL, PO1_CAD RC_Mark, ME1_CAD RC_Mark, M1_CAD Dummy_Block, PO1_CAD Dummy_Block and DIFF_CAD Dummy_Block for the devices:NCAP_11_SP, NCAP_18_SP, NCAP_25_SP	

Ver.	Phase	Approved Date	From	To	Remark (Purpose)
B02	B	11/12/2010	-	8. Add CDF parameter of nwell option:RNPP0_SP, RNNPO_SP 9. Modified Logic/MM MOSFET Pcells : 9-1 Condition of "Length=60nm, Fingers=1, Finger Width $\leq 0.13\mu\text{m}$" DRC rule DF.A issue. 9-2 Small "w" (as know as dogbone) abutted, the MOS devices did not align well. 9-3 DRC violation (M1-CT.EN2.RCM) appeared when MOS finger width is set at 0.155um, 0.165um, 0.175um, 0.185um and DFM on:N_11_SPSHVT, P_11_SPSHVT, N_11_SPRVT, P_11_SPRVT, N_11_SPLVT, P_11_SPLVT, N_11_SPHVT, P_11_SPHVT. 10. Modified DIOP_SP, DION_SP Pcell for least one contact. 11. Technology file design rules are restricting some Metal Minimum widths to 0.2um. The design rule manual allows minimum widths of 0.1u. 12. Add/update c equation for NCAP_11_SP, NCAP_18_SP, NCAP_25_SP, MOMCAPS_SY_MMKF and MOMCAPS_AS_MMKF. 13. Extend N_25_SP, P_25_SP maximum length range to 50um 14. Extend DION_SP and DIOP_SP range, $0.5u \leq W \leq 1.5u$ to $0.5u \leq W \leq 20u$, $20u \leq L \leq 50u$ to $0.5u \leq L \leq 50u$ 15. Modify CDF termMapping of compact model's MOS, BJT, diode hspiceD view 16. Upgrade callback re-trigger, loadProcess, loadVoltage 17. Add Assura DRC, LVS & QRC 18. Updated to current version of model files, technology file and command files.	

Ver.	Phase	Approved Date	From	To	Remark (Purpose)
B03	B	06/11/2014	-	1. Change or add Metal options base on newly TLR:1P6M0T1H0U_option74 to 1P6M0T1H0U_option351P7M0T1H0U_option76 to 1P7M0T1H0U_option391P8M0T2H0U_option77 to 1P8M0T2H0U_option431P8M0T1H0U_option78 to 1P8M0T1H0U_option831P9M0T2H0U_option41 to 1P9M0T2H0U_option88 1P5M1T0F0U_option2, 1P6M2T0F0U_option4, 1P6M1T1F0U_option5, 1P7M1T0F0U_option7, 1P7M2T0F0U_option8, 1P7M2T1F0U_option10, 1P7M0T2F0U_option11, 1P7M0T1F0U_option12, 1P8M2T0F0U_option13, 1P8M2T2F0U_option16, 1P6M1T1F1U_option65, 1P9M1T1F1U_option71, 1P8M1T0F0U_option82, 1P4M0T1H0U_option32, 1P5M1T1H0U_option33, 1P6M1T1H0U_option34, 1P7M1T1H0U_option36, 1P8M1T1H0U_option40, 1P6M1T2H0U_option75, 1P10M2T1F1U_option31, 1P8M0T1H0U_option83, 1P5M0T1F0U_option51, 1P7M0T2H0U_option38, 1P6M1T0F0U_option3, 1P6M0T1H0U_option35, 1P7M0T1H0U_option39, 1P8M0T1H0U_option83, and 1P8M0T2H0U_option43 2. Add Chapter.7 OSS-based AMS Netlister Application 3. Modify the Chapter.1 Metal Options	

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